TUTORIAL 6

Vector Spaces, Subspaces, and Linear Combinations

MOTIVATION AND GUIDING IDEA

Objectives. We will formulate problems as linear systems, perform row reduction while preserving the solution set, and distinguish free variables from pivot variables. We will connect homogeneous systems with vector spaces and test closure systematically.

NOTATION AND CONVENTIONS

If $A \in M_{m\times n}(F)$, then $Ax = b$ is a system of m equations in n variables, with $x \in F^n$ and $b \in F^m$. The notation $[A \mid b]$ denotes its augmented matrix, and $A \sim B$ denotes row equivalence. The rank rank A is the number of pivots. Define $\ker A = \{x \in F^n : Ax = 0\}$. The expression $\operatorname{span}\{v_1, \dots, v_k\}$ denotes all combinations $\alpha_1 v_1 + \cdots + \alpha_k v_k$ with $\alpha_j \in F$. The notation $x_0 + U = \{x_0 + u : u \in U\}$ denotes a translate of a subspace U. Distinguish the empty set $\varnothing$, with no elements, from $\{0\}$, which has one element.

DEFINITION 6.1

Vector space axioms and solution tools. A vector space over a field F is a set V equipped with operations $V \times V \to V$ and $F \times V \to V$ satisfying, for all $u, v, w \in V$ and $\alpha, \beta \in F$,

$$u + v = v + u, \qquad (u + v) + w = u + (v + w).$$

There is $0_V \in V$ such that $v + 0_V = v$, and every $v \in V$ has an additive inverse $-v \in V$ satisfying $v + (-v) = 0_V$. In addition,

$$\alpha(u + v) = \alpha u + \alpha v, \qquad (\alpha + \beta)v = \alpha v + \beta v,$$

$$(\alpha\beta)v = \alpha(\beta v), \qquad 1_F v = v.$$

For $A \in M_{m\times n}(F)$, the system $Ax = b$ is consistent if and only if $\operatorname{rank} A = \operatorname{rank}[A \mid b]$. If x_0 is a particular solution, then its full solution set is $x_0 + \ker A$: indeed, $Ax = b$ is equivalent to $A(x - x_0) = 0$. This solution set is a subspace of F^n if and only if $b = 0$, since containing the zero vector forces $A0 = b = 0$.

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6.1 Vector spaces

RECALL AND TOOLS

1. To define a vector space V over a field $\mathbb{F}$, we specify an addition of vectors in V and a multiplication of a vector in V by a scalar in $\mathbb{F}$, satisfying the vector space axioms. These are **in addition** to addition and multiplication in $\mathbb{F}$. A product of two vectors is not part of the vector space axioms; if needed later, it will be defined separately.
2. Every field is a vector space over itself, with the field's addition and multiplication.
3. A vector space V is always defined over a field $\mathbb{F}$. Always specify the field of scalars. For example, as discussed in the lecture, $\mathbb{C}$ can be regarded as a vector space over $\mathbb{C}$ or over $\mathbb{R}$. These are two different vector spaces.

EXAMPLE 6.1

Let $\mathbb{F}$ be a field.

(a) $\mathbb{F}^n$: the set of n-tuples over $\mathbb{F}$, where $n \in \mathbb{N}$.

(b) $M_{m\times n}(\mathbb{F})$: the set of $m \times n$ matrices over $\mathbb{F}$.

(c) $\mathbb{F}[X]$: the set of all polynomials with coefficients in $\mathbb{F}$.

EXERCISE 6.1

Please determine whether the following sets are subspaces of $\mathbb{R}[X]$ over $\mathbb{R}$:

1. $V = \{p(X) \in \mathbb{R}[X] : p(-2) = 0,\ p(3) \neq 0\}$.
2. $V = \{p(X) \in \mathbb{R}[X] : \exists c \in \mathbb{R} \setminus \{0\},\ p(c) = 0\}$.

WORKED SOLUTION

1. No. The zero polynomial is absent. Addition also fails: for
$$p(X) = (X+2)(X-8), \qquad q(X) = (X+2)^2,$$
we have $p(-2) = q(-2) = 0$ and $p(3) = -25$, $q(3) = 25$, but
$$(p+q)(X) = (X+2)(2X-6), \qquad (p+q)(3) = 0.$$
2. No. The polynomials $(X-2)^2$ and $(X+2)^2$ belong to the set, with roots $2, -2$, respectively. Their sum is $2X^2 + 8 > 0$ for all real X, so it does not belong.

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6.2 Subspaces

DEFINITION 6.2

Let V be a vector space over a field $\mathbb{F}$, and let $U \subseteq V$ be a subset that is itself a vector space over $\mathbb{F}$ under the addition and scalar multiplication inherited from V. Then U is a vector subspace of V.

PROPOSITION 6.1

Let V be a vector space over a field $\mathbb{F}$, and let $U \subseteq V$ be a subset. Then U is a vector subspace if and only if all of the following conditions hold:

1. $0 \in U$, where this is the zero vector of V.
2. U is closed under addition.
3. U is closed under scalar multiplication.

EXAMPLE 6.2

1. The set $V = \{p(X) \in F[X] : p(-3) = 0\}$ is a subspace of $F[X]$. It contains the zero polynomial. For $p, q \in V$ and $\alpha \in F$,

$$(p+q)(-3) = p(-3) + q(-3) = 0, \qquad (\alpha p)(-3) = \alpha p(-3) = 0.$$

   Thus it is closed under addition and scalar multiplication.

2. For $n \geq 3$, the set $W = \{x \in \mathbb{R}^n : x_1 + x_3 = 0\}$ is a subspace. It contains zero. If $x, y \in W$, then

$$(x_1 + y_1) + (x_3 + y_3) = (x_1 + x_3) + (y_1 + y_3) = 0.$$

   Also $(\alpha x)_1 + (\alpha x)_3 = \alpha(x_1 + x_3) = 0$ for every $\alpha \in \mathbb{R}$.

EXERCISE 6.2

Please prove or please disprove:

1. The set $U = \{(a, 3a^2) : a \in \mathbb{R}\}$ is a subspace of $\mathbb{R}^2$.
2. The symmetric matrices form a subspace of $M_n(F)$.

WORKED SOLUTION

1. This is not a subspace: $(1, 3) \in U$, but

$$(1,3) + (1,3) = (2,6) \notin U,$$

   because a first coordinate of 2 requires second coordinate $3 \cdot 2^2 = 12$.

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EXERCISE 6.2 CONTINUED

2. The zero matrix is symmetric. If $A^T = A$, $B^T = B$ and $\lambda \in F$, then

$$(A+B)^T = A^T + B^T = A + B, \qquad (\lambda A)^T = \lambda A^T = \lambda A.$$

Thus all three subspace-test conditions hold.

6.3 Vector Spaces

RECALL AND TOOLS

1. To define a vector space V over a field $\mathbb{F}$, specify addition of vectors in V and multiplication of a vector in V by a scalar in $\mathbb{F}$, satisfying the vector-space axioms. These operations are specified **in addition** to the addition and multiplication in $\mathbb{F}$. Multiplication of two vectors is not a vector-space axiom; any such operation used later must be defined separately.
2. Every field is a vector space over itself, using its own addition and multiplication.
3. A vector space V is always defined over a field $\mathbb{F}$, which must be stated. For example, as discussed in the lecture, $\mathbb{C}$ can be viewed as a vector space over $\mathbb{C}$ or over $\mathbb{R}$. These are two different vector-space structures.

EXAMPLE 6.3

Let $\mathbb{F}$ be a field.

(a) $\mathbb{F}^n$: the set of n-tuples over $\mathbb{F}$ ($n \in \mathbb{N}$).

(b) $M_{m\times n}(\mathbb{F})$: the set of $m \times n$ matrices over $\mathbb{F}$.

(c) $\mathbb{F}[X]$: the set of all polynomials with coefficients in $\mathbb{F}$.

EXERCISE 6.3

Please determine whether the following sets are subspaces of $\mathbb{R}[X]$ over $\mathbb{R}$ with the usual operations.

1. $V = \{p(X) \in \mathbb{R}[X] : p(-3) = 0,\ p(2) \neq 0\}$.
2. $W = \{p(X) \in \mathbb{R}[X] : \exists c \in [0,1],\ p(c) = 0\}$.

WORKED SOLUTION

1. No. The zero polynomial does not belong to V. Addition also fails: for

$$p(X) = (X+3)(X-1), \qquad q(X) = -(X+3),$$

we have $p(-3) = q(-3) = 0$, $p(2) = 5$, and $q(2) = -5$. Thus $p, q \in V$, but

$$(p+q)(X) = (X+3)(X-2), \qquad (p+q)(2) = 0,$$

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EXERCISE 6.3 CONTINUED

so $p+q \notin V$.

2. No. Both $p(X) = X$ and $q(X) = 1 - X$ belong to W, with roots 0 and 1. Their sum is the constant polynomial 1, which has no root in the interval. Thus W is not closed under addition.

6.3.1 Subspaces

DEFINITION 6.3

Let V be a vector space over a field $\mathbb{F}$. A subset $U \subseteq V$ is a **vector subspace** of V if it is itself a vector space over $\mathbb{F}$ under the addition and scalar multiplication inherited from V.

PROPOSITION 6.2

Let V be a vector space over a field $\mathbb{F}$ and $U \subseteq V$. Then U is a subspace if and only if all the following conditions hold:

1. $0 \in U$, where 0 is the zero vector of V.
2. U is closed under addition.
3. U is closed under scalar multiplication.

EXAMPLE 6.4

1. $V = \{p(X) \in \mathbb{F}[X] : p(-3) = 0\}$.
2. $U = \{x \in \mathbb{R}^n : x_1 = 2x_2\}$, where $n \geq 2$.

Both sets are subspaces. The zero polynomial lies in V, and for $p, q \in V$ and $\alpha \in \mathbb{F}$,

$$(p+q)(-3) = p(-3) + q(-3) = 0, \qquad (\alpha p)(-3) = \alpha p(-3) = 0.$$

The zero vector also lies in U. If $x, y \in U$ and $\alpha \in \mathbb{R}$, then

$$(x+y)_1 = x_1 + y_1 = 2x_2 + 2y_2 = 2(x+y)_2,$$

$$(\alpha x)_1 = \alpha x_1 = 2\alpha x_2 = 2(\alpha x)_2.$$

Thus both sets are closed under addition and scalar multiplication.

EXERCISE 6.4

Please prove or disprove.

1. $U = \{(a, 5a^2)^T : a \in \mathbb{R}\}$ is a subspace of $\mathbb{R}^2$.
2. The symmetric matrices form a subspace of $M_n(\mathbb{F})$.

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EXERCISE 6.4 CONTINUED

WORKED SOLUTION

1. No.
$$(1,5)^T \in U, \qquad (1,5)^T + (1,5)^T = (2,10)^T \notin U,$$
because $5 \cdot 2^2 = 20 \neq 10$, so addition is not closed.
2. Yes. $0^T = 0$, and if $A^T = A$, $B^T = B$, and $\lambda \in \mathbb{F}$, then
$$(A+B)^T = A^T + B^T = A + B, \qquad (\lambda A)^T = \lambda A^T = \lambda A.$$
All three subspace conditions hold.

Additional practice

6.4 Linear Combinations

DEFINITION 6.4

Let V be a vector space over a field $\mathbb{F}$ and let $v_1, \ldots, v_k \in V$. A vector $v \in V$ is a **linear combination** of $v_1, \ldots, v_k$ if there are $a_1, \ldots, a_k \in \mathbb{F}$ such that $v = a_1 v_1 + \cdots + a_k v_k$.

EXERCISE 6.5

Please determine whether A lies in the span of B_1, B_2 over $\mathbb{R}$.
$$A = \begin{pmatrix} -2 & 4 \\ 3 & -3 \end{pmatrix}, \quad B_1 = \begin{pmatrix} 1 & -1 \\ 0 & 0 \end{pmatrix}, \quad B_2 = \begin{pmatrix} 0 & 0 \\ 1 & -1 \end{pmatrix}.$$

WORKED SOLUTION

$$aB_1 + bB_2 = \begin{pmatrix} a & -a \\ b & -b \end{pmatrix}.$$
Equality with A would require $a = -2$ and $-a = 4$, hence $a = -4$. This contradiction proves that A is not in the span. The choice $b = 3$ matches only the second row.

EXERCISE 6.6

Let $A = \begin{pmatrix} 1 & -2 \\ 0 & 0 \end{pmatrix}$ over $\mathbb{R}$. Please characterize all matrices B satisfying $AB = BA$, and please show that they are precisely the linear combinations of A and $I_2 - A$.

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EXERCISE 6.6 CONTINUED

WORKED SOLUTION

Write $B = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$. Computing both products gives

$$AB = \begin{pmatrix} a-2c & b-2d \\ 0 & 0 \end{pmatrix}, \qquad BA = \begin{pmatrix} a & -2a \\ c & -2c \end{pmatrix}.$$

Equality requires $c = 0$ and $b = -2(a-d)$. Hence

$$B = \begin{pmatrix} a & -2(a-d) \\ 0 & d \end{pmatrix} = aA + d(I_2 - A).$$

Substitution into both products shows conversely that every matrix of this form commutes with A.

6.4.1 Spanning Sets

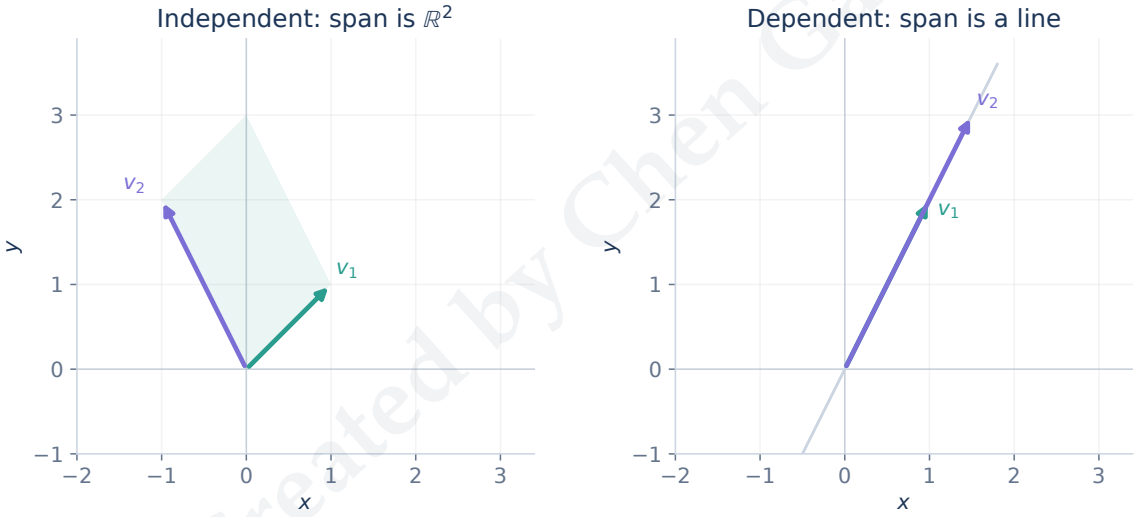


Figure 6.1. Geometric span: a nonzero vector spans a line through the origin; two independent vectors span a plane through the origin.

DEFINITION 6.5

Let V be a vector space over a field $\mathbb{F}$. For a set $S \subseteq V$, not necessarily finite, define $\operatorname{span}_{\mathbb{F}} S$ as the set of all finite linear combinations of elements of S:

$$\operatorname{span}_{\mathbb{F}} S = \left\{ \sum_{j=1}^{k} \alpha_j v_j \;\middle|\; k \geq 0,\ v_j \in S,\ \alpha_j \in \mathbb{F} \right\}.$$

The empty sum ($k = 0$) is allowed and equals the zero vector. In particular, $\operatorname{span}_{\mathbb{F}} \varnothing = \{0\}$.

NOTE

If V is a vector space over $\mathbb{F}$ and $S \subseteq V$, then span S is a subspace of V.

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DEFINITION 6.6

If V is a vector space over $\mathbb{F}$ and $S \subseteq V$ satisfies $V = \operatorname{span} S$, then S **spans** V and is called a **spanning set**. If S is finite, V is **finitely generated**.

NOTE

Pay attention to the scalar field. A set $S \subseteq V$ can span over one field and fail to span over another. When more than one field is possible, write $\operatorname{span}_{\mathbb{F}} S$ to specify the field.

EXERCISE 6.7

Let V be a vector space over $\mathbb{F}$, let $\varnothing \neq S \subseteq V$ be a finite set of size k, and let $v \in V$. Please prove that

$$\operatorname{span}(S \cup \{v\}) = \operatorname{span} S \quad \Longleftrightarrow \quad v \in \operatorname{span} S.$$

WORKED SOLUTION

$\Longrightarrow$: Assume $\operatorname{span}(S \cup \{v\}) = \operatorname{span} S$. Write $v = 0 \cdot v_1 + \cdots + 0 \cdot v_k + 1 \cdot v$. Thus v is a linear combination of elements of $S \cup \{v\}$, so $v \in \operatorname{span}(S \cup \{v\})$. The assumed equality gives $v \in \operatorname{span} S$.

$\Longleftarrow$: The inclusion $\operatorname{span}(S \cup \{v\}) \supseteq \operatorname{span} S$ always holds: any linear combination of elements of S is also one of elements of $S \cup \{v\}$, with coefficient zero on v. For the opposite inclusion, write $S = \{s_1, \ldots, s_k\}$ and take $x \in \operatorname{span}(S \cup \{v\})$. There are $a_1, \ldots, a_{k+1} \in \mathbb{F}$ with

$$a_1 s_1 + \cdots + a_k s_k + a_{k+1} v = x.$$

Since $v \in \operatorname{span} S$, there are $b_1, \ldots, b_k \in \mathbb{F}$ with $b_1 s_1 + \cdots + b_k s_k = v$. Substituting gives

$$(a_1 + a_{k+1} b_1) s_1 + \cdots + (a_k + a_{k+1} b_k) s_k = x.$$

Hence $x \in \operatorname{span} S$, as required.

EXERCISE 6.8

Please find a spanning set for each solution space over $\mathbb{R}$.

1. $\begin{cases} x + 2y = 0, \\ x - z = 0. \end{cases}$
2. $2y - z = 0.$

WORKED SOLUTION

1. Setting $y = t$ gives $x = -2t$ and $z = x = -2t$. Hence

$$S_1 = \{t(-2, 1, -2) : t \in \mathbb{R}\} = \operatorname{span}\{(-2, 1, -2)\}.$$

Substitution verifies that each multiple satisfies both equations.

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EXERCISE 6.8 CONTINUED

2. Set $x = s$, $y = t$; then $z = 2t$. Thus

$$(x, y, z) = s(1, 0, 0) + t(0, 1, 2), \qquad S_2 = \text{span}\{(1, 0, 0), (0, 1, 2)\}.$$

Both generators satisfy the equation, and the representation includes every solution.

6.5 Intersections and Sums of Subspaces

MOTIVATION AND GUIDING IDEA

Distinguish union from sum. For example, the two coordinate axes in $\mathbb{R}^2$ are subspaces, but their union is not closed under addition: $(2, 0)$ and $(0, -3)$ belong to the union, whereas their sum $(2, -3)$ does not. The sum of the two axes is the whole plane.

RECALL AND TOOLS

If $U, W \subseteq V$ are subspaces of a vector space V, their **intersection** is

$$U \cap W = \{v \in V : v \in U \text{ and } v \in W\}.$$

DEFINITION 6.7

For subspaces $U, W \subseteq V$, define their **sum** by

$$U + W = \{u + w : u \in U, w \in W\}.$$

Then $U + W$ is a subspace of V containing both U and W. If also $U \cap W = \{0\}$, the sum is called a **direct sum** and is denoted $U \oplus W$.

EXERCISE 6.9

Please write

$$\text{Sym}(\mathbb{F}) = \{A \in M_n(\mathbb{F}) : A = A^T\}, \qquad \text{aSym}(\mathbb{F}) = \{A \in M_n(\mathbb{F}) : A = -A^T\}$$

for the symmetric and skew-symmetric matrices. Please prove that both sets are subspaces of $M_n(\mathbb{F})$. In addition, please prove that if $\text{char}(\mathbb{F}) \neq 2$, that is, $1 + 1 \neq 0$ in the field, then

$$M_n(\mathbb{F}) = \text{Sym}(\mathbb{F}) \oplus \text{aSym}(\mathbb{F}).$$

WORKED SOLUTION

We verify that $\text{Sym}(\mathbb{F})$ is a subspace. For $\text{aSym}(\mathbb{F})$, the zero matrix belongs to the set, and if $A^T = -A$, $B^T = -B$, and $\lambda \in \mathbb{F}$, then

$$(A + B)^T = A^T + B^T = -A - B = -(A + B), \qquad (\lambda A)^T = \lambda A^T = -\lambda A.$$

Thus skew-symmetric matrices are also closed under addition and scalar multiplication and form a subspace.

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EXERCISE 6.9 CONTINUED

Now let $A, B \in \text{Sym}(\mathbb{F})$ and $\lambda \in \mathbb{F}$. First, $0 \in \text{Sym}(\mathbb{F})$. For addition, $(A+B)^T = A^T + B^T = A + B$, so $A + B$ is symmetric. For scalar multiplication, $(\lambda A)^T = \lambda A^T = \lambda A$, so λA is symmetric. To prove the direct-sum assertion, first prove that the sum is the whole matrix space and then prove that the intersection is zero. For the equality $M_n(\mathbb{F}) = \text{Sym}(\mathbb{F}) + \text{aSym}(\mathbb{F})$, check both inclusions:

1. $M_n(\mathbb{F}) \supseteq \text{Sym}(\mathbb{F}) + \text{aSym}(\mathbb{F})$ holds because both summands are subspaces of $M_n(\mathbb{F})$.
2. For the reverse inclusion, take $A \in M_n(\mathbb{F})$ and write

$$A = \frac{A + A^T}{2} + \frac{A - A^T}{2}.$$

The first summand is symmetric, since

$$\left(\frac{A + A^T}{2}\right)^T = \frac{A^T + (A^T)^T}{2} = \frac{A + A^T}{2},$$

and the second is skew-symmetric, since

$$\left(\frac{A - A^T}{2}\right)^T = \frac{A^T - (A^T)^T}{2} = \frac{A^T - A}{2} = -\frac{A - A^T}{2}.$$

Hence $M_n(\mathbb{F}) \subseteq \text{Sym}(\mathbb{F}) + \text{aSym}(\mathbb{F})$.

Finally, suppose $A \in \text{Sym}(\mathbb{F}) \cap \text{aSym}(\mathbb{F})$. Symmetry gives $A = A^T$, while skew-symmetry gives $-A = A^T$. Thus $A = -A$, and $a_{ij} = -a_{ij}$ for every $1 \le i, j \le n$. Consequently $a_{ij} + a_{ij} = 2a_{ij} = 0$. Because $\text{char}(\mathbb{F}) \neq 2$, the scalar $2 = 1 + 1$ is nonzero and invertible in the field. Therefore every $a_{ij} = 0$, so $A = 0$, as required.

RECALL AND TOOLS

In characteristic 2, $-A = A$, so the conditions $A^T = A$ and $A^T = -A$ coincide. This is exactly why the symmetric/skew-symmetric decomposition using division by 2 requires characteristic different from 2.

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6.6 Intersections and Sums of Subspaces

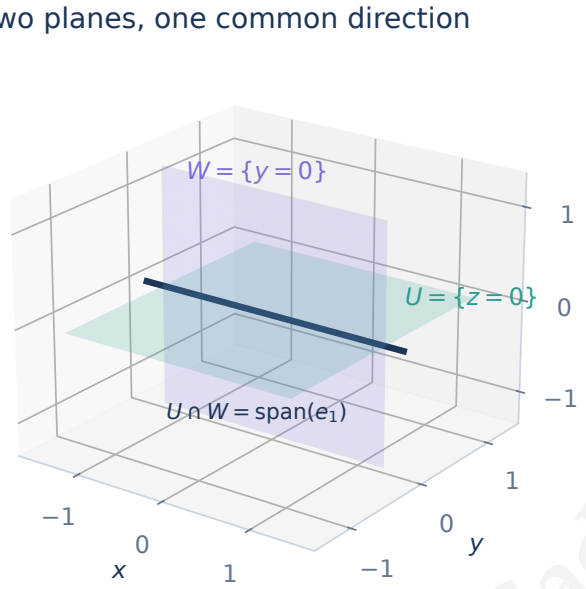


Figure 6.2. Two distinct planes through the origin in $\mathbb{R}^3$ may intersect in a line. Here $U = \{z = 0\}$ and $W = \{y = 0\}$, so $U \cap W = \text{span}\{(1, 0, 0)\}$ and $U + W = \mathbb{R}^3$.

RECALL AND TOOLS

For subspaces $U, W \subseteq V$ of a vector space V, their intersection is

$$U \cap W = \{v \in V : v \in U \text{ and } v \in W\}.$$

DEFINITION 6.8

For subspaces $U, W \subseteq V$, define

$$U + W = \{u + w : u \in U, w \in W\}.$$

This is a subspace of V containing U and W. If also $U \cap W = \{0\}$, the sum is a **direct sum**, denoted $U \oplus W$.

EXERCISE 6.10

Let

$$\text{Sym}(\mathbb{F}) = \{A \in M_n(\mathbb{F}) : A = A^T\}, \qquad \text{aSym}(\mathbb{F}) = \{A \in M_n(\mathbb{F}) : A = -A^T\}.$$

Please prove that the symmetric and skew-symmetric matrices are subspaces of $M_n(\mathbb{F})$, and that whenever $\text{char}(\mathbb{F}) \neq 2$,

$$M_n(\mathbb{F}) = \text{Sym}(\mathbb{F}) \oplus \text{aSym}(\mathbb{F}).$$

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EXERCISE 6.10 CONTINUED

WORKED SOLUTION

We show that $\mathrm{Sym}(\mathbb{F})$ is a subspace. For the skew-symmetric set, the corresponding identities are

$$0^T = -0, \quad (A+B)^T = -A - B = -(A+B), \quad (\lambda A)^T = -\lambda A,$$

so this set is also a subspace. For the symmetric set, take $A, B \in \mathrm{Sym}(\mathbb{F})$ and $\lambda \in \mathbb{F}$. Clearly $0 \in \mathrm{Sym}(\mathbb{F})$. Closure under addition follows from $(A+B)^T = A^T + B^T = A + B$. Closure under scalar multiplication follows from $(\lambda A)^T = \lambda A^T = \lambda A$. To establish the direct sum, first prove equality with the sum and then show that the intersection is zero. Check both inclusions for $M_n(\mathbb{F}) = \mathrm{Sym}(\mathbb{F}) + \mathrm{aSym}(\mathbb{F})$:

1. $M_n(\mathbb{F}) \supseteq \mathrm{Sym}(\mathbb{F}) + \mathrm{aSym}(\mathbb{F})$ follows because both summands are subspaces of $M_n(\mathbb{F})$.
2. Take $A \in M_n(\mathbb{F})$. Then
$$A = \frac{A + A^T}{2} + \frac{A - A^T}{2}.$$
The first summand is symmetric because
$$\left(\frac{A + A^T}{2}\right)^T = \frac{A^T + (A^T)^T}{2} = \frac{A + A^T}{2}.$$
The second is skew-symmetric because
$$\left(\frac{A - A^T}{2}\right)^T = \frac{A^T - (A^T)^T}{2} = \frac{A^T - A}{2} = -\frac{A - A^T}{2}.$$
This proves the reverse inclusion.

Now take $A \in \mathrm{Sym}(\mathbb{F}) \cap \mathrm{aSym}(\mathbb{F})$. Symmetry gives $A = A^T$, and skew-symmetry gives $-A = A^T$. Therefore $A = -A$, so $a_{ij} = -a_{ij}$ for every $1 \le i, j \le n$. Thus $a_{ij} + a_{ij} = 2a_{ij} = 0$. As the field has characteristic different from 2, the scalar 2 is nonzero, so every $a_{ij} = 0$. Hence $A = 0$, as required.

EXERCISE 6.11

$$D = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}, \qquad H = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}.$$

Please compute DH and HD entry by entry, please compare their action on e_2, and please find all v satisfying $DHv = HDv$.

WORKED SOLUTION

$$DH = \begin{pmatrix} 2\cdot 1 + 0\cdot 0 & 2(-1) + 0\cdot 1 \\ 0\cdot 1 + 1\cdot 0 & 0(-1) + 1\cdot 1 \end{pmatrix} = \begin{pmatrix} 2 & -2 \\ 0 & 1 \end{pmatrix},$$

$$HD = \begin{pmatrix} 1\cdot 2 + (-1)0 & 1\cdot 0 + (-1)1 \\ 0\cdot 2 + 1\cdot 0 & 0\cdot 0 + 1\cdot 1 \end{pmatrix} = \begin{pmatrix} 2 & -1 \\ 0 & 1 \end{pmatrix}.$$

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EXERCISE 6.11 CONTINUED

Thus $DHe_2 = (-2,1)^T$, whereas $HDe_2 = (-1,1)^T$. For $v = (x,y)^T$,

$$DHv - HDv = (2x - 2y - (2x - y),\ y - y)^T = (-y, 0)^T.$$

The results agree exactly when $y = 0$, on $\mathrm{span}\{e_1\}$.

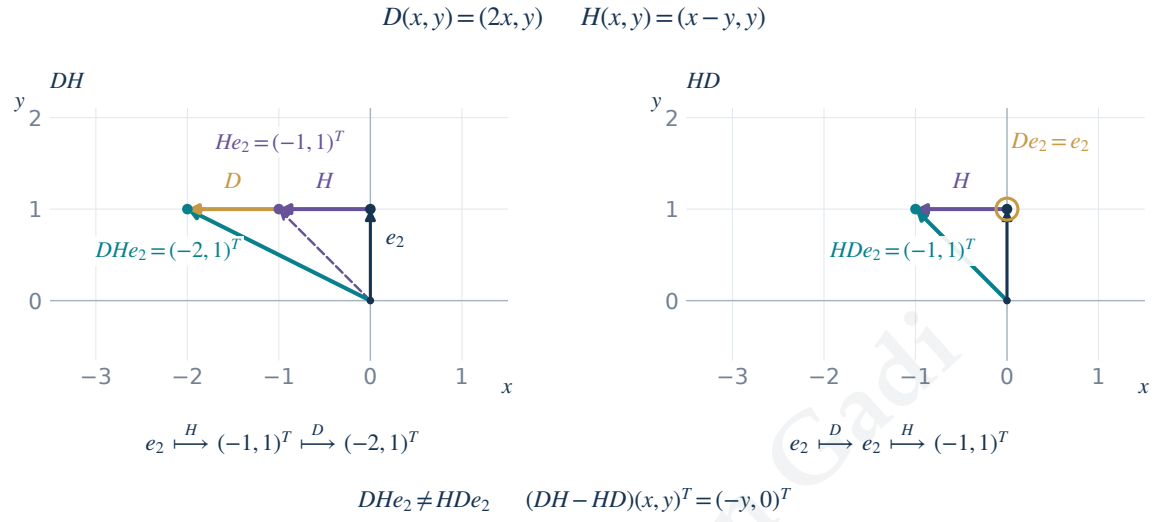


Figure 6.3. Operation order on the same vector: DH applies H first, whereas HD applies D first. The ring marks the fixed point $De_2 = e_2$, but the final results differ: $DHe_2 = (-2,1)^T \neq (-1,1)^T = HDe_2$.

EXERCISE 6.12

Let V be a vector space over a field $\mathbb{F}$, and let $S, R \subseteq V$ be arbitrary subsets, not necessarily finite.
Please write

$$\mathrm{span}(S) = \left\{ \sum_{i=1}^{k} \alpha_i s_i \;\middle|\; k \in \mathbb{Z}_{\geq 0},\ \alpha_i \in \mathbb{F},\ s_i \in S \right\},$$

The empty sum is the zero vector, so $\mathrm{span}(\varnothing) = \{0\}$. Define $\mathrm{span}(R)$ and $\mathrm{span}(S \cup R)$ similarly.
Please show that, for all $S, R \subseteq V$,

$$\mathrm{span}(S) + \mathrm{span}(R) = \mathrm{span}(S \cup R).$$

WORKED SOLUTION

Write

$$U := \mathrm{span}(S), \qquad W := \mathrm{span}(R).$$

Recall that the sum $U + W$ is defined by

$$U + W := \{u + w \mid u \in U,\ w \in W\}.$$

We prove

$$\mathrm{span}(S) + \mathrm{span}(R) = \mathrm{span}(S \cup R)$$

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EXERCISE 6.12 CONTINUED

by establishing both inclusions:

$$\text{span}(S) + \text{span}(R) \subseteq \text{span}(S \cup R) \quad \text{and} \quad \text{span}(S \cup R) \subseteq \text{span}(S) + \text{span}(R).$$

1. First prove
$$\text{span}(S) + \text{span}(R) \subseteq \text{span}(S \cup R).$$
Let $v \in \text{span}(S) + \text{span}(R)$. Then $v = u + w$ for some $u \in \text{span}(S)$ and $w \in \text{span}(R)$:
$$v = u + w.$$
By the definition of $\text{span}(S)$, there are vectors $s_1, \ldots, s_k \in S$ and scalars $\alpha_1, \ldots, \alpha_k \in \mathbb{F}$ such that
$$u = \sum_{i=1}^{k} \alpha_i s_i.$$
Similarly, there are vectors $r_1, \ldots, r_\ell \in R$ and scalars $\beta_1, \ldots, \beta_\ell \in \mathbb{F}$ such that
$$w = \sum_{j=1}^{\ell} \beta_j r_j.$$
Therefore,
$$v = u + w = \sum_{i=1}^{k} \alpha_i s_i + \sum_{j=1}^{\ell} \beta_j r_j.$$
Every s_i and every r_j lies in $S \cup R$. Thus v is a finite linear combination of vectors in $S \cup R$.
By the definition of span,
$$v \in \text{span}(S \cup R).$$
Hence
$$\text{span}(S) + \text{span}(R) \subseteq \text{span}(S \cup R).$$
2. For the reverse inclusion, prove
$$\text{span}(S \cup R) \subseteq \text{span}(S) + \text{span}(R).$$
Let $v \in \text{span}(S \cup R)$. There are $x_1, \ldots, x_t \in S \cup R$ and $\gamma_1, \ldots, \gamma_t \in \mathbb{F}$ such that
$$v = \sum_{i=1}^{t} \gamma_i x_i.$$
Partition the indices into disjoint sets: assign to S every vector belonging to S, and assign the rest to R. A vector in the intersection is counted only in the first set.
Define
$$I_S := \{i \in \{1, \ldots, t\} \mid x_i \in S\}, \qquad I_R := \{1, \ldots, t\} \setminus I_S.$$
Then v can be written as
$$v = \sum_{i \in I_S} \gamma_i x_i + \sum_{i \in I_R} \gamma_i x_i.$$

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EXERCISE 6.12 CONTINUED

Set

$$u := \sum_{i \in I_S} \gamma_i x_i, \qquad w := \sum_{i \in I_R} \gamma_i x_i.$$

Observe that

- for $i \in I_S$, $x_i \in S$, so u is a linear combination of vectors in S; thus

$$u \in \text{span}(S).$$

- for $i \in I_R$, $x_i \in R$, so w is a linear combination of vectors in R; thus

$$w \in \text{span}(R).$$

We have obtained

$$v = u + w$$

with $u \in \text{span}(S)$ and $w \in \text{span}(R)$, hence

$$v \in \text{span}(S) + \text{span}(R).$$

This proves

$$\text{span}(S \cup R) \subseteq \text{span}(S) + \text{span}(R).$$

The two inclusions give

$$\text{span}(S) + \text{span}(R) = \text{span}(S \cup R),$$

as required.

EXERCISE 6.13 FOUNDATION

Please solve over $\mathbb{R}$ the system

$$\begin{cases} x + y + z = 6, \\ x - y + z = 2. \end{cases}$$

Please write its solution set as a particular solution plus multiples of a fixed vector. Please determine whether it is a subspace and please describe all differences between pairs of solutions.

WORKED SOLUTION

Subtracting the second equation from the first gives $2y = 4$, so $y = 2$ and $x + z = 4$. Put $z = t$ to get

$$S = \{(4 - t, 2, t) : t \in \mathbb{R}\} = (4, 2, 0) + \text{span}\{(-1, 0, 1)\}.$$

It is not a subspace, because zero does not satisfy the equations. The difference of two solutions is

$$(4 - t, 2, t) - (4 - s, 2, s) = (t - s)(-1, 0, 1).$$

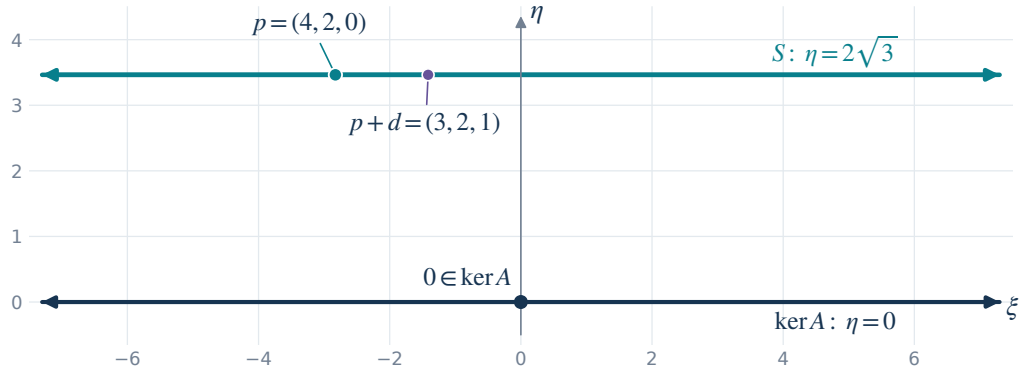
Every real multiple occurs by choosing $s = 0$. Thus the differences form exactly $\text{span}\{(-1, 0, 1)\}$, the solution set of the homogeneous system.

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$$S=\{(4-t,2,t):t\in\mathbb{R}\}=p+\ker A$$

$$\Phi(\xi,\eta)=\xi u+\eta v,\qquad u=\frac{(-1,0,1)}{\sqrt{2}},\quad v=\frac{(1,1,1)}{\sqrt{3}}$$



$$\ker A=\operatorname{span}\{d\},\quad d=(-1,0,1),\qquad 0\notin S$$

Figure 6.4. The real system $x+y+z=6$, $x-y+z=2$ has solution set $S=p+\ker A$, where $p=(4,2,0)$ and $\ker A=\operatorname{span}\{(-1,0,1)\}$. The ξ,η axes describe the plane $\operatorname{span}\{u,v\}\subseteq\mathbb{R}^3$ through the map Φ displayed in the figure. The vectors u,v are perpendicular unit vectors, so the chart preserves lengths and angles. The translated line excludes the origin, whereas the kernel contains it.

EXERCISE 6.14 CHALLENGE

Let $a,b\in\mathbb{R}$. Please classify the solutions of

$$\begin{cases}x+y+z=5,\\ x+ay+z=b,\\ x+y+az=b.\end{cases}$$

In each case, please find the coefficient matrix rank, the number of free variables when consistent, and the solution set. Also find the kernel of the coefficient matrix, and please determine whether the solution set is a subspace.

WORKED SOLUTION

Subtract the first equation from both the second and the third. The equivalent system is

$$\begin{cases}x+y+z=5,\\ (a-1)y=b-5,\\ (a-1)z=b-5.\end{cases}$$

1. If $a\neq 1$, division by $a-1$ is permitted. Put $c=(b-5)/(a-1)$. The unique solution is

   $$S=\{(5-2c,c,c)\}.$$

   There are three pivots, so the rank is 3 and there are no free variables. In the homogeneous system, $y=z=0$ and then $x=0$, so $\ker A=\{0\}$.

2. If $a=1$ and $b=5$, the last two equations after subtraction are $0=0$. The rank is 1, and

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EXERCISE 6.14 CHALLENGE CONTINUED

there are two free variables. Choosing $y = s, z = t$ gives

$$S = \{(5 - s - t, s, t) : s, t \in \mathbb{R}\} = (5, 0, 0) + \text{span}\{(-1, 1, 0), (-1, 0, 1)\}.$$

The kernel is $\text{span}\{(-1, 1, 0), (-1, 0, 1)\}$.

3. If $a = 1$ and $b \neq 5$, the equation $0 = b - 5$ is a contradiction, so $S = \varnothing$. The coefficient matrix still has rank 1, while the augmented matrix has rank 2. The kernel depends only on A, so it is the same kernel as in the preceding case.

The solution set is never a subspace: in the consistent cases the zero vector fails to satisfy $x + y + z = 5$, and in the inconsistent case the set is empty. A number of degrees of freedom of a solution set cannot be defined through a choice of solution parameters when no solutions exist.

EXERCISE 6.15 CHALLENGE

Let U, W be subspaces of a vector space V over an arbitrary field $\mathbb{F}$. Please prove that the following are equivalent:

1. $U \cup W$ is a subspace of V.
2. $U \subseteq W$ or $W \subseteq U$.
3. $U + W = U \cup W$.

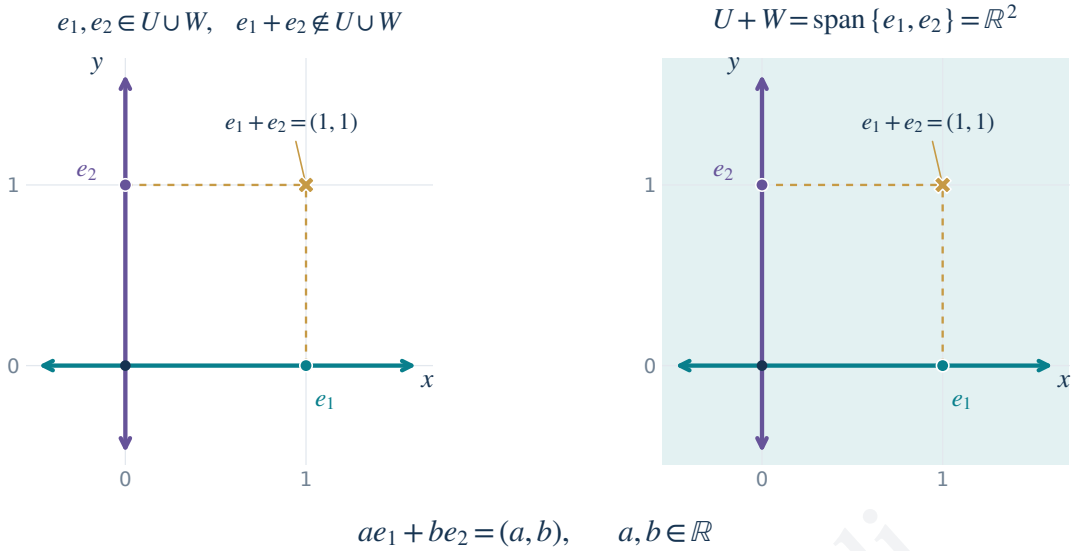
WORKED SOLUTION

(1) $\Rightarrow$ (2): Suppose neither inclusion holds. Choose $u \in U \setminus W$ and $w \in W \setminus U$. Both belong to the union, so closure under addition gives $u + w \in U \cup W$. If $u + w \in U$, then $w = (u + w) - u \in U$, a contradiction. If $u + w \in W$, then $u = (u + w) - w \in W$, again a contradiction. Hence one inclusion must hold.
(2) $\Rightarrow$ (3): If $U \subseteq W$, then $U \cup W = W$ and $U + W = W$: every such sum belongs to W, and each $w \in W$ equals $0_V + w$. The other case follows by interchanging the subspaces.
(3) $\Rightarrow$ (1): The sum $U + W$ is a subspace, so the equal set $U \cup W$ is also a subspace. The proof divides by no scalar and therefore applies to finite fields and characteristic 2 as well.

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$$U = \mathbb{R}e_1, \qquad W = \mathbb{R}e_2, \qquad e_1 = (1,0), \quad e_2 = (0,1)$$



$$ae_1 + be_2 = (a,b), \qquad a,b \in \mathbb{R}$$

Figure 6.5. An illustration over $\mathbb{R}$: the coordinate axes $U = \mathbb{R}e_1$ and $W = \mathbb{R}e_2$ are subspaces. Their union contains e_1 and e_2 but excludes their sum $(1,1)$, so it is not closed under addition. In contrast, every $(a,b) \in \mathbb{R}^2$ equals $ae_1 + be_2$, giving $U + W = \mathbb{R}^2$. Shading in the right panel represents the full plane within the viewing window.

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EXERCISE 6.16 CHALLENGE

Let $S \subseteq \mathbb{R}^n$ be nonempty and assume $(1-t)x + ty \in S$ for all $x, y \in S$ and $t \in \mathbb{R}$.

1. Please choose $a \in S$ and prove that $U = \{x - a : x \in S\}$ is a subspace and $S = a + U$, where $a + U = \{a + u : u \in U\}$.
2. Is it enough to require only $(x + y)/2 \in S$ for all $x, y \in S$? Please prove or disprove.

WORKED SOLUTION

We have $0 = a - a \in U$. For $u \in U$ and $\lambda \in \mathbb{R}$, the hypothesis gives

$$(1 - \lambda)a + \lambda(a + u) = a + \lambda u \in S,$$

so $\lambda u \in U$. For $u, v \in U$, taking the midpoint of $a + u$ and $a + v$ gives $(u + v)/2 \in U$. Scalar closure with scalar 2 then gives $u + v \in U$. Hence U is a subspace. Its definition gives both inclusions in $S = a + U$. Midpoint closure alone is insufficient. The set $\mathbb{Q} \subseteq \mathbb{R}$ is closed under midpoints. If it were $a + U$ for a real subspace U, the fact that $0 \in \mathbb{Q}$ would imply $\mathbb{Q} = U$. But $1 \in U$ would then imply $\sqrt{3} \cdot 1 \in U$, a contradiction.

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EXERCISE 6.17 CHALLENGE

Let $U_1, \ldots, U_m$ be proper subspaces of a real vector space V, meaning $U_j \neq V$ for every j.

1. Please prove that $U_1 \cup \cdots \cup U_m \neq V$.
2. Please construct four proper subspaces of $\mathbb{Z}_3^2$ whose union is the whole space.

WORKED SOLUTION

We induct on m. The case $m = 1$ is the assumption. By the induction hypothesis, choose

$$a \notin U_1 \cup \cdots \cup U_{m-1}, \qquad b \notin U_m.$$

Consider $a + tb$ for $t \in \mathbb{R}$. For each $j < m$, at most one value of t gives a point in U_j. Two distinct values would imply $b \in U_j$ by subtraction and division by their difference, and then $a \in U_j$, a contradiction. The same line meets U_m in at most one point, since two would imply $b \in U_m$. Only finitely many parameters are excluded, whereas $\mathbb{R}$ is infinite. This proves the claim. Over $\mathbb{Z}_3$, take

$$U_1 = \operatorname{span}\{(1,0)\}, \quad U_2 = \operatorname{span}\{(0,1)\},$$
$$U_3 = \operatorname{span}\{(1,1)\}, \quad U_4 = \operatorname{span}\{(1,2)\}.$$

Each subspace has three vectors, so each is proper. A vector (x, y) with $x = 0$ belongs to U_2. If $x \neq 0$, then $(x, y) = x(1, y/x)$, where y/x is 0, 1, or 2, so the vector lies in U_1, U_3, or U_4. The union covers all nine vectors.

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EXERCISE 6.18 CHALLENGE

Let $a_1, \ldots, a_{n+2} \in \mathbb{R}^n$, where $n \geq 1$. For a nonempty index set I, define the convex hull by

$$\operatorname{conv}\{a_i : i \in I\} = \left\{ \sum_{i \in I} \lambda_i a_i : \lambda_i \geq 0,\ \sum_{i \in I} \lambda_i = 1 \right\}.$$

Please prove that the indices can be partitioned into two nonempty sets I, J whose convex hulls intersect.

WORKED SOLUTION

Consider real coefficients $\alpha_1, \ldots, \alpha_{n+2}$ satisfying

$$\sum_{i=1}^{n+2} \alpha_i a_i = 0, \qquad \sum_{i=1}^{n+2} \alpha_i = 0.$$

This is a homogeneous system of $n+1$ equations in $n+2$ unknowns, so it has a nonzero solution. Since the coefficients sum to zero, at least one is positive and at least one is negative. Put

$$I = \{i : \alpha_i \geq 0\}, \qquad J = \{i : \alpha_i < 0\}, \qquad s = \sum_{i \in I} \alpha_i = -\sum_{j \in J} \alpha_j > 0.$$

These sets are nonempty and partition all indices. The vector equation yields

$$\sum_{i \in I} \frac{\alpha_i}{s} a_i = \sum_{j \in J} \frac{-\alpha_j}{s} a_j.$$

On each side the coefficients are nonnegative and sum to 1. The common point therefore lies in both convex hulls.

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